Zurich ServiceNow AI Platform Capabilities

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Application services

Understand application services, learn about different application service types and how multiple ServiceNow® business units and products use them.

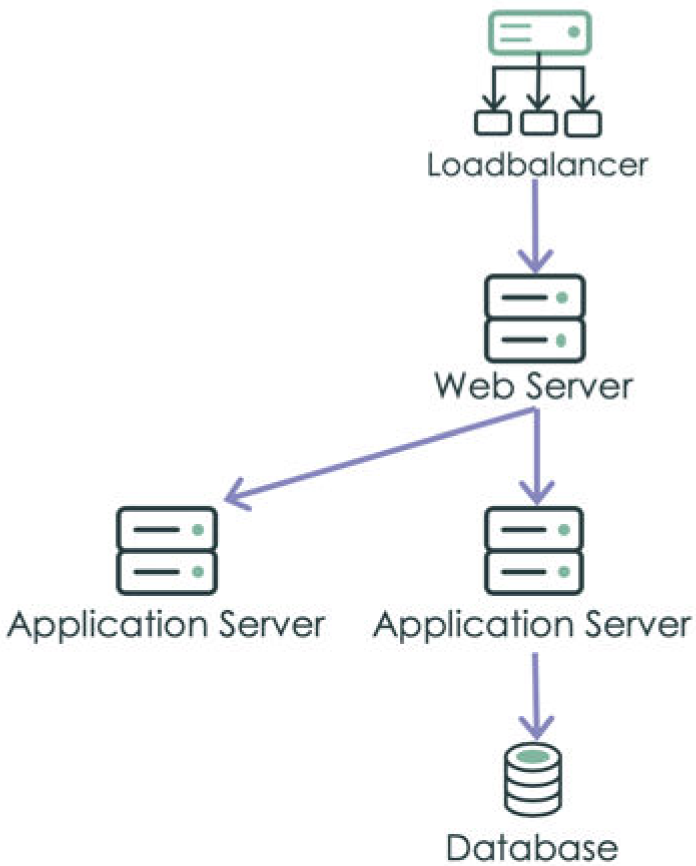
What application services are

A service instance is a set of interconnected applications and hosts that are configured to offer a service to the organization. Service instances can be internal, like an organization email system or customer-facing, like an organization website. For example, creating financial reports through a web-based application requires a computer, web server, application server, databases, middleware, and network infrastructure. These applications and hosts are all configured to offer the service of financial reporting. In development environments, an application service represents an instance of a business application or system.

ServiceNow applications refer to devices and applications that comprise an application service as configuration items (CIs). The various CIs and the relationships between them, that comprise an application service, are stored in the Configuration Management Database (CMDB).

Each application service contains an entry point as the top-level CI. An entry point is a point where clients access a service instance. Typically, it is a URL, or a combination of the IP address and port for application services in enterprise deployments. For cloud-based deployments, an entry point can be a URL to a cloud resource like an AWS gateway.

Application service



The Common Service Data Model (CSDM) helps you streamline service types and service offerings. You can add relationships between application services and other service-related objects in the CSDM: Business Application, Technical Service Offerings, or Business Service Offerings.

There are the following types of application services:

Discovered

Service Mapping discovers application services using patterns and by following traffic connections.

Pattern-based discovery creates precise and complete application services that represent the service-centric view of the IT infrastructure. It creates a high-fidelity map that is well suited to managing mission-critical application services.

In addition, it provides visibility of cloud-native services such as compute, load balancers, and API gateways. You can use service entry points such as AWS S3 buckets, AWS and Microsoft Azure API gateways, AWS Lambda functions, and Microsoft Azure functions to map services. It can also detect Lambda to Lambda calls and Lambda to RDS connections to build dynamic service maps.

Top-down method maps VMs on-premises and in public clouds. However, it requires these VMs to be fully discovered for the top-down discovery to determine which applications are running in the VM. If a VM isn't fully discovered, use the tag-based method to bridge the gap (see later in this document). Tag-based mapping also maps containers, that you cannot map using the top-down discovery.

Discovered application services have the service classification of application service. They are stored in the Mapped Application Service [cmdb_ci_service_discovered] table.

Dynamic CI Group

Dynamic CI groups which act as application services. The members of the CMDB groups that is associated with the dynamic CI group, populates the application service. A dynamic CI group is a dynamic grouping of CIs, based on some common criteria such as the location of all web servers in Detroit or all Oracle databases in Boston. After creating a dynamic CI group, it can be used as a group offering in IT Service Management.

If created from the Application Service wizard, the service classification is application service, and if created from the legacy Event Management UI or Service Mapping UI, the classification is technical service. Application services of the Dynamic CI Group type are stored in the Dynamic CI Group [cmdb_ci_query_based_service] table.

Tag-based

A tag is a label that consists of a key-value pair. Your organization may use tags to categorize its assets, to enhance query and reporting capabilities. Discovery and Cloud Provisioning and Governance can discover tags used by all major cloud providers and container ecosystems. Once the tags are discovered, Service Mapping can create service instances based on these tags. For example, you can use tags to map all application services your organization uses in the production environment in the EMEA region.

Tag-based application services have the service classification of application service. They are stored in the Tag-based Application Service [cmdb_ci_service_by_tags] table.

Created Manually

With manual mapping, application owners manually document the applications, IT infrastructure, and relationships that support each application service. This methodology is the best fit for configuration items that are not fully discoverable due to security access issues. For example, IPS devices which support an intrusion prevention service for the security business unit.

Try to avoid manual mapping wherever possible. It’s incredibly time consuming to map services manually, and often the information needed for mapping is not available due to evolving technology and a lack of processes that track and document the infrastructure dependencies needed for application context. And, whenever subsequent changes are made to the application service topology, the service map must be manually updated.

Manually created application services have the service classification of application service. Application services of the created manually type are stored in the Mapped Application Service [cmdb_ci_service_discovered] table.

Dynamic

A dynamic application service includes only CIs that are part of CI relationships stored in the CMDB CI Relationship [cmdb_rel_ci] table.

You can't edit a dynamic application service by directly adding or removing CIs from it. Dynamic application services are updated automatically to reflect any change to CI relationships in the CMDB CI Relationship [cmdb_rel_ci] table. When you add a relationship to a CI that is contained in a dynamic application service, then that service automatically updates to reflect the addition of the relationship and the associated new CI. In a similar manner, a dynamic application service automatically updates upon the removal of a relationship and its associated CI from a CI within the service.

One way to create dynamic application services, is by converting legacy business services or legacy manual services (created with Event Management, for example) into application services of the dynamic type.

Dynamic application services have the service classification of application service. Dynamic application services are stored in Calculated Application Services [cmdb_ci_service_calculated] table.

Who uses application services

Application services provide foundation for operation of the following business units and products of the ServiceNow AI Platform:

- ITOM Health gathers alerts from infrastructure events captured by third-party monitoring tools. It then uses IT-related information gathered by Discovery to map alerts to configuration items. Based on the collected information, then provides dashboards showing a consolidated view of all service-impact events.
- ITOM Optimization gives you tools to provision private and public cloud infrastructure and services and to achieve consistent management and cost visibility. The Cloud Cost Management application, available in the ServiceNow Store, helps you to analyze the full range of costs associated with cloud assets so you can identify and take action on opportunities to save money and optimize operations.
- IT Service Management users rely on the application services reflecting the IT infrastructure to manage and deliver services to their customers.
- Customer Service Management users efficiently diagnose and resolve issues related to the IT infrastructure in the context of application services.
- Software Asset Management users understand the software running in your IT environment and track configurations that impact software license consumption across your IT environments and datacenters.
- Strategic Portfolio Management users utilize data collected for application services to gain a comprehensive understanding of the applications used in your organization.

How to create application services

Depending on the needs of your organization, you can deploy different methods of creating and populating application services.

> **Important:** You can use the top-down and manual methods for the same application service. You cannot combine any other methods for creating or populating the same application service.

Analyze the IT infrastructure and service deployment in your organization to pick the optimal method of creating and populating application services.

Choosing the right method for your deployment

Method	When to use	Additional considerations
Top-down discovery Service Mapping performs top-down discovery of application services. Service Mapping uses patterns to discover and map CIs. A pattern is a sequence of steps whose purpose is to detect attributes of a CI and its outbound connections. This method creates precise and complete application services that reliably represent the service-aware view of your organization's IT infrastructure Tag-based discovery in Service Mapping	Use this method to discover industry-recognized or customized second-tier and third-tier applications. Such applications may include load-balancing solutions, application or web servers with database connections.	Pattern-based mapping requires configuring credentials, users, and user permissions to let Service Mapping access applications inside your organization private network. This process may take time and effort.

Method	When to use	Additional considerations
is a complimentary method that enriches the results of top-down discovery.		
Tag-based If your organization uses tags for asset management, you can use these tags to map application services. Discovery and Cloud Provisioning and Governance discover tags assigned to CIs, and populate the CMDB with this data. Service Mapping uses the tag-related data from the CMDB to map services. Tag-based service mapping complements top-down service mapping. It provides visibility of containers and also maps VMs that aren't fully discovered, which top-down service mapping is unable to do. However, while tag-based mapping associates tagged components with specific application services, it doesn't	Map resources on cloud workloads like IaaS/Paas/FaaS/CaaS as well as on container workloads using Kubernetes, OpenShift, or AWS ECS. Also, map resources in the Site Reliability Engineering (SRE) or Customer Reliability Engineering (CRE) deployments. Using tag-based method, you can map container resources in your deployments. Typically, you use this method to discover applications on cloud virtualizations or PaaS deployments.	Unlike other mapping methods, tag-based mapping doesn't require configuring credentials or providing users with elevated rights. Tag-based application services may not include relevant CIs, if these CIs don't have correct tags assigned to them.

Method	When to use	Additional considerations
map the connections between these components—This is another reason why tag-based mapping complements rather than replaces top-down service mapping.		
Ingesting Application Performance Management (APM) maps from integrated Dynatrace or AppDynamics deployments Create application services using the integration with AppDynamics application model and Dynatrace monitoring platform available on ServiceNow Store.	Use this integration to create application services based on APM maps from Dynatrace or AppDynamics. You are able to use application services created by this method for monitoring Health.	Analyze discovered resources in the CMDB before ingesting from 3rd party to avoid creating duplicate CIs.
Populate an application service using the Dynamic CI Group method Based on CMDB groups, whose members populate the application service.	Use this method as a simple and fast way to create dynamic CI groups for deployments including Microsoft Active Directory, Microsoft Exchange or other DNS-related services. Dynamic CI Groups are especially useful if only a list of resource	There is no map view for application services created using this method. You can only view CIs belonging to such an application service as a list. Need to ensure that the CMDB group

Method	When to use	Additional considerations
	is available, without configuration details or credentials. Using a CMDB group lets you use CMDB Health to monitor health, and use a CMDB Query Builder saved query to filter for the CIs included in the application service.	accurately filters for the CIs that should be included in the application service.
Application service API Create an automation for creating application services in bulk. Use this method, if your organization has performed cross-organization mapping and analysis and collected some information about services. Application services created using APIs belong to the manual type are stored in the Mapped Application Service [cmdb_ci_service_discovered] table.	Use this method for environments that require tracing of DevOps Continuous Integration/ Continuous Deployment (CI/CD) process. You can import third-party service maps into manual application services individually or in bulk. For example, see the Digital Guidebook: Importing 3rd-party service maps into ServiceNow Service Mapping.	Be familiar with the exact service structure: sys_id of each CI comprising the service and the hierarchy that the CIs form. This method requires knowledge of the scripting infrastructure that your organization uses.
Populate an application service using the Manual method	Use the manual method if you can't use other methods of	This method doesn't require any preexisting setup or object configuration.

Method	When to use	Additional considerations
Create a manual application service with one CI only: the entry point. To populate a manually created application service, add other CIs manually as described in Manually add CIs to an application service. Alternatively, create and populate manual application services by converting business services created in the CMDB and stored in [cmdb_ci_service].	creating or populating application services. Create application services manually for intrusion prevention.	You can include CIs of any class in manually created application services. Manually created application services reflect some changes to CIs, like CI attributes. However, they do not automatically reflect changes to CI relationships.
Populate an application service using the Dynamic Service method Application services that automatically update to reflect any change to CI relationships in the CMDB CI Relationship [cmdb_rel_ci] table. To conform with Common Service Data Model, you can also convert legacy services to dynamic application services. Those legacy services are stored in the [cmdb_ci_service] or	Use this method to transform legacy business services into application services that other ServiceNow products can utilize. For example, dynamic application services can be used for service monitoring and change management.	You can't edit a dynamic application service by adding or removing CIs from it. The system automatically modifies an application service of the dynamic type when you modify relevant relationships for CIs that are part of that application service.

Method	When to use	Additional considerations
[cmdb_ci_service_manual] CMDB tables: • Convert business services to application services • Convert legacy manual services into dynamic application services		
From CSV file Service Mapping extracts information from this file and creates potential application services referred to as service candidates. Use this method, if your organization has performed cross-organization mapping and analysis and collected some information about services.	If necessary, you can import service candidates from multiple CSV files.	Organize all the collected information in a specific order in a CSV file, precisely as described in the documentation.

To comply with CSDM, convert manual services created using IT Operations Management Event Management and stored in [cmdb_ci_service_manual] as covered in Convert manual services to application services or Convert manual services to application services using API. Converted services become application services of the manual type stored in the Mapped Application Service [cmdb_ci_service_discovered] table.

Domain separation

Domain separation, if deployed, impacts an service instance as follows:

- When creating an service instance, the service instance is assigned to the user's domain.
- When manually adding a CI to an service instance, you can choose only CIs that belong to the service domain.
- When using the createOrUpdateService - POST REST API for creating or updating an application service, the process stops if one of the CIs referenced in the API belongs to a different domain than the application service itself.
- When converting business services into application services, the newly created application service belongs to the same domain as the original business service. The application service comprises only CIs belonging to the same domain as the application service itself.

Create an application service

Create an application service to adhere to Common Service Data Model standards and to standardize the organization, maintenance, and monitoring of services in your organization.

Before you begin

Role required: Depending on the population method used:

- Dynamic CI Group: app_service_admin
- Manual: app_service_admin
- Dynamic Service: app_service_admin
- Top-Down Discovery: sm_admin
- Tags: sm_admin

About this task

An application service is a set of interconnected applications and hosts, which are configured to offer a service for the organization. Application services can be internal, like an organization's email system, or customer-facing, like an organization's website.

An application service has an entry point, which lets users access the application service. If you are at the planning stage and don't know what the entry points are for an application service, you can create the application service without entry points. Such an application service is referred to as an empty application service, to which you can add entry points at any later time.

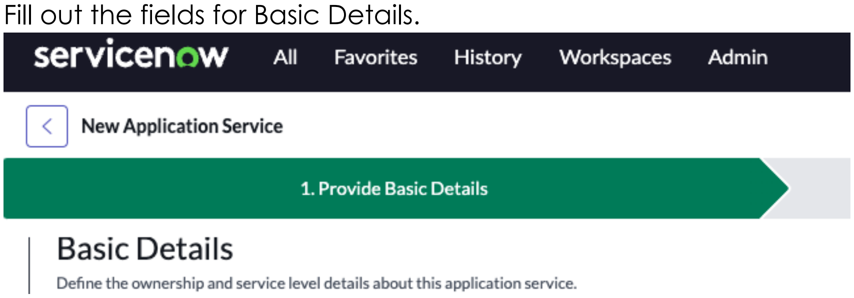
All application services created in the Application Service wizard, are set with the application service classification.

Service Mapping, if activated, can automatically discover and map application services as described in Application service mapping. A discovered application service contains the CIs and the connections between them that Service Mapping discovered and mapped.

You can also create an application service by using the createOrUpdateService - POST REST API.

Procedure

1. Navigate to **All > CSDM > Manage Technology Management Services > Service Instance**.
2. In the Application Services list view, select **New** to open the Application Service wizard.
3. In the **Provide Basic Details** tab:
   a. Fill out the fields for Basic Details.



Field	Description
Number	Pre-populated unique ID for the application service.
Name	Unique application service name, which isn't in use by any other type of application service. Use self-explanatory names such as `mailing service` or `printing service`.
Environment	The environment of the offering such as production, development, or test, as identified by some service offerings. Used by Incident Management and Change Management.
Version	The application service configuration version.
Model ID	A product model such as a software model where end of life data is stored.
Operational Status	Operational status of the application service, such as **Ready** or **Retired**.
Support Group	Used by Incident Management as the group managing the contract covering the asset.

Field	Description
Change Group	Used by ITSM for routing of change and change-related tasks.
Managed By Group	Group responsible for managing the data.
Owned By	User who is familiar with the infrastructure and applications making up the service. This user is the application service Subject Matter Expert (SME) who provides information necessary for a successful creation of an application service. If the owner name is not listed, create a user with the sm_app_owner role, as the owner. Alternatively, you can choose a user with the sm_admin role.

Note: See Teams related list for details about the automatic synchronization between the assignment group fields and the Teams related list.

b. In the Set Relationships section, add relationships between the application service and other components in the CSDM domain.

c. Select the **Business Application**, **Technical Service Offering**, **Business Service Offering**, or the **Parent Application Service** tab, and add items to the respective **Selected** list.

- The technical service offering list includes records in the Service Offering [service_offering] table, in which the Service classification attribute is **Technical Service**.

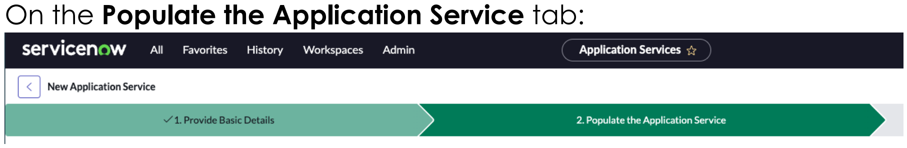
- The business service offering list includes records in the Service Offering [service_offering] table, in which the Service classification attribute is **Business Service**.
- The parent application service list includes application service records from the Service Instance [cmdb_ci_service_auto] table. Adding a parent application service relationship creates hierarchies and dependencies of application services in deployments such as:
    - Platform host and platform application deployments
    - Micro service deployments in which one or more micro services identified as an application service, is part of a larger application service deployment
    - Shared technical service dependencies

d. Select **Next**.

For information about CSDM relationships, see CI relationships in the CSDM.

Also, some fields and relationships are noted as required on the page. To change which fields and which relationships are required, see Modify the attributes and relationships required for application services.

4. On the **Populate the Application Service** tab:



a. Click **Choose a Method** or select **Next** to skip selecting a service population method.

b. On the Choose a Method page, select a **Service Population Method**, and then follow the respective link to complete the specific population method:

- **Top Down Discovery**: Use top-down discovery to populate application services

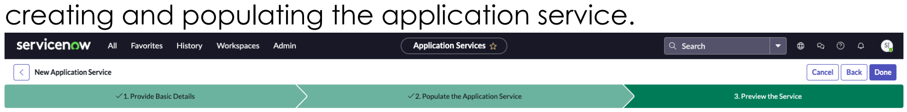
- **Dynamic CI Group**: Populate an application service using the Dynamic CI Group method.
- **Tags**: Use tags to populate application services
- **Manual**: Use the Manual method to populate application services
- **Dynamic Service**: Populate an application service using the Dynamic Service method

  Note: The **Top Down Discovery** and the **Tags** options are available only if Service Mapping is installed.

c. To add another method to populate the application service, click **Add Method** on the Service Population Methods page. Or, select **Next**.

- You can add any combination of the **Top Down Discovery** and the **Manual** methods. However, if you select the **Dynamic CI Group**, **Tags**, or the **Dynamic Service** method, the **Add Method** button is grayed out and you can't add additional methods.
- You can select a card for a Converted Business Service method to see details about the service conversion, such as the conversion type. For more information, see Convert business services to application services.

5. On the **Preview the Service** tab, review and verify the summaries for creating and populating the application service.



a. Review **Relationships**.

b. You can select **Edit Relationships** to modify the relationships to other CSDM objects.

c. Review **Population Methods Summary**.

d. You can select **Edit Methods** to modify the selection methods.

e. Select **Done**.

Result

The application service is created, and you can access the new application service by navigating to application services list views:

- **CSDM > Manage Technology Management Services > Application Service**: Contains application service CIs from any class extending the Service Instance [cmdb_ci_service_auto] class, except alert groups. The list view includes the tag-based, discovered, manual, dynamic CI groups, converted, dynamic, and empty application service types.
- **Configuration > Application Services > Application Services**: Contains application service CIs from any class extending the Service Instance [cmdb_ci_service_auto] class, except alert groups. The list view includes the tag-based, discovered, manual, dynamic CI groups, converted, dynamic, and empty application service types.
- **Service Mapping > Services > Service Instances**: Contains application service CIs from the Mapped Application Service [cmdb_ci_service_discovered] class. The list view includes the top-down (discovered) and empty application service types.

What to do next

- If the service population method is **Dynamic CI Group**:
  - Select **View CMDB Group CI's** to list all the CIs in the CMDB group that is associated with the application service.
  - Select **View Service CI's** to list all the CIs in the application service.

  Both lists of CIs are identical, unless the CMDB group contains more than 10,000 CIs. In this case, **View CMDB Group CI's** shows all the CIs in the CMDB group, and **View Service CI's** shows only the 10,000 CIs that are members of the application service.
- If the service population method is **Tags**, **Top Down Discovery**, or **Manual**, and select **View Map** to view the application service map where you can:
  - Link application services
  - View CI attributes in an application service map
  - View the change history of application services

- Compare two versions of an application service

- Select **Advanced**, and then on the Advanced Details page, select **Additional Info**, **Questionnaire**, **Reject Messages**, or **Worknotes**, to add details.

Use top-down discovery to populate application services

Use top-down discovery to populate an application service. This discovery method deploys discovery patterns to find configuration items (CIs) belonging to the service and connections between these CIs. Pattern-based discovery creates precise and complete service instances that reliably represent the service-aware view of your organization's IT infrastructure.

Before you begin

- Top-Down Discovery is one of several methods for populating an application service with CIs. Choosing a method for populating an application service, is only one step of the generic procedure for creating an application service. Ensure that you have completed the initial steps as described in Create an application service. The procedure described here is incomplete by itself as it complements that generic procedure.
- Verify that Service Mapping is set up properly.
- Ensure you know which entry point to use for this application service and which attributes you must be able to define for this entry point. Learn about Entry point attributes available with Service Mapping.

For information about the different types of application services and the different methods you can use to populate application services, including using top-down discovery, see Application services.

Role required: sm_admin

About this task

A pattern is a sequence of commands designed to detect attributes of a CI and its outbound connections. Service Mapping and Discovery share a set of preconfigured patterns that cover most of the commonly used devices and applications.

Service Mapping starts pattern-based top-down discovery process from the entry point you define.

An entry point is a point where clients access a service instance. Usually, it is either a URL or a combination of the IP address and port. Service Mapping starts the mapping process from this point. For example, to map your electronic mailing service instance, define an IP address or host name of the email server as an entry point.

Entry points vary depending on the nature of the service instance. Service Mapping comes with a wide range of preconfigured entry point types that cover many commonly used applications.

Procedure

1. From the **Service Population Method** list in the Choose a Method window, select **Top-Down Discovery**.
2. From the **Application Type** list, select the CI class of the application that serves as the entry point for this application service.
   Entry point parameters depend on the type you select.
3. Define attributes for the selected entry point as described in Entry point attributes.
4. (Optional) Add free-text comment that may provide useful information for handling this application service later.
5. Click **Save**.

What to do next

Complete the generic procedure Create an application service.

Related topics

- Discovery patterns used by ITOM Visibility

Use the Dynamic CI Group method to populate application services

The Dynamic CI Group method for populating an application service, automatically generates a dynamic CI group. The members of the CMDB group that the dynamic CI group is based on, populates the application service. The application service continuously synchronizes with the CMDB group to reflect any changes in membership in the CMDB group.

Before you begin

The Dynamic CI Group is one of several methods for populating an application service with CIs. Choosing a method for populating an application service, is only one step of the generic procedure for creating an application service. Ensure that you have completed the initial steps as described in Create an application service. The procedure described here is incomplete by itself as it complements that generic procedure.

Note:

- The number of CIs in an application service that is populated by the Dynamic CI Group method, is limited to 10,000, even if the associated CMDB group has more than 10,000 CIs.
- A CMDB group can be used to populate only a single application service. For more information about populating and using CMDB groups, see CMDB groups.
- A dynamic CI group contains CIs but can't contain other groups.

For information about the different types of application services and the different methods you can use to populate application services, including Dynamic CI Group, see Application services.

Role required: app_service_admin

Procedure

1. In the Choose a Method page, select **Dynamic CI Group** as the **Service Population Method**.
2. Fill out the fields that appear, which are specific to the Dynamic CI Group service population method.

Field	Description
Service Population Method	The method used for populating the application service with CIs. Set to **Dynamic CI Group**.
CMDB Table	Notes the Dynamic CI Group [cmdb_ci_query_based_service] table, in which application services created by the Dynamic CI Group method, are stored.
Group Name	The CMDB group whose members become members the application service. **Note:** CIs from a class that extends the cmdb_ci_service class (Services), are automatically filtered out and are not added to the application service.

3. Select **Save**.

Result

The alert impact on dynamic CI groups is calculated on the following CIs:

- All CIs that are part of the dynamic CI's CMDB group.
- Children of current CIs with a relationship of: Runs on::Runs
- CIs related to either the current CIs or their children, with a relationship of: Virtualized by::Virtualizes

For more information, see alert impact calculation.

What to do next

Complete the generic procedure Create an application service.

Use tags to populate application services

Use tags that help categorize and organize configuration items (CIs) in your organization to populate an service instance. Tag-based mapping doesn't require configuring credentials or providing users with elevated rights. Tag-based population method requires the Service Mapping feature of ITOM Visibility.

Before you begin

1. Tags is one of several methods for populating an application service with CIs. Choosing a method for populating an application service, is only one step of the generic procedure for creating an application service. Ensure that you have completed the initial steps as described in Create an application service. The procedure described here is incomplete by itself as it complements that generic procedure.

2. Analyze the tag usage in your organization and make a list of all tags and their purposes. Use the Key Value [cmdb_key_value] table to see the tags in the CMDB.

3. If necessary, assign tags to CIs that you want to include in an service instance.

Role required: sm_admin

About this task

A tag is a label that consists of a key-value pair. Your organization may use tags to categorize its assets, to enhance query and reporting capabilities. Discovery and Cloud Provisioning and Governance can discover tags used by all major cloud providers and container ecosystems. Once the tags are discovered, Service Mapping can create service instances based on these tags. For example, you can use tags to map all application services your organization uses in the production environment in the Europe, the Middle East and Africa (EMEA) region.

If you have configured tag-based service families and tag categories, you can use these tag definitions for populating an application service. Part of defining a tag-based service family is defining a tag category, which contains tag keys. If necessary, you can also define tag values to narrow the criteria used for populating application services. Based on the tag definitions for the tag-based service family, Service Mapping creates service candidates - suggested application services. When you use the tag-based service families to populate an application service, you must select the relevant service candidate.

Alternatively, you can define tag keys and their values while choosing the tag-based population method for a new application service. Define up to three tag keys and tag values for the population criteria. CIs that have discovered tag keys and tag values, become part of an service instance.

For information about the different types of application services and the different methods you can use to populate application services, including using tags, see Application services.

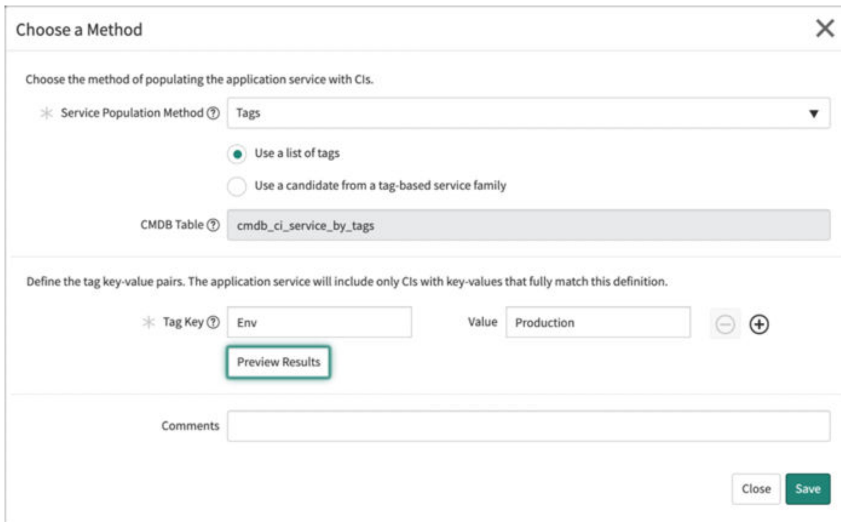
> **Note:** Service Mapping includes CIs that are part of CI relationships even if these CIs do not have tags assigned to them. For more information, see Tag-based discovery in Service Mapping.

Procedure

1. From the **Service Population Method** list in the Choose a Method window, select **Tags**.
2. To define new tag criteria, perform the following steps:
   a. Select **Use a list of tags**.
   b. Enter the tag key and its respective tag value.
      Matching tag keys that exist in the system, appear in the auto-fill options.

      > **Important:** Tag-based mapping is not case-sensitive; same key names and key values spelled with upper and lower case are identified as the same.

   c. (Optional) Click the plus icon and add another tag key and tag value.



d. Click **Preview Result** to see the list of CIs that match the defined criteria.

> **Note:** If necessary, refine the tag definitions. You can add no more than three tag key-value pairs for one application service.

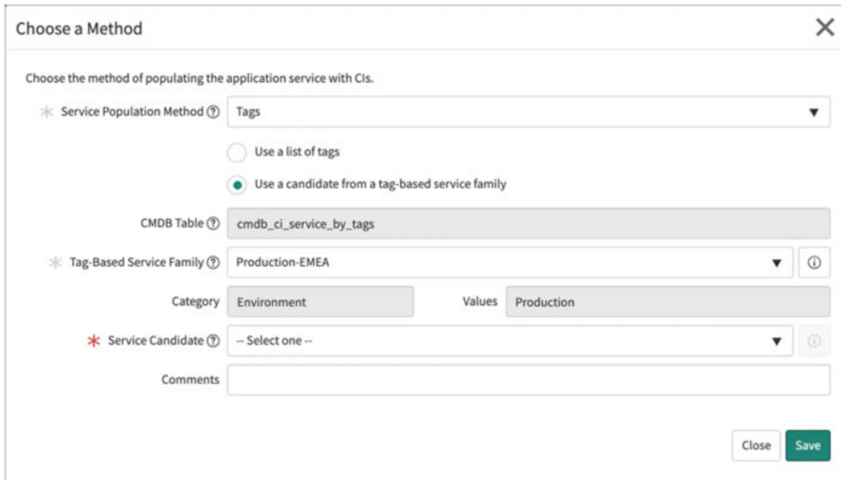
3. To use tag definitions from a preconfigured tag-based service family, perform the following steps:

   a. Select **Use a candidate from a tag-based service family**.

   b. (Optional) To see the tag definitions for this tag-based service family, click the **Preview** button 

   c. From the **Tag-Based Service Family** list, select the relevant family.

   d. Review the tag categories and values assigned to the service family.



e. From the **Service Candidate** list, select the relevant candidate.

f. (Optional) To review the service candidate form, click the **Preview** button .

4. (Optional) Add free-text comment that may provide useful information for handling this application service later.

5. Click **Save**.

What to do next

Complete the generic procedure Create an application service.

Use the Manual method to populate application services

The Manual method for populating an application service, is based on selecting an entry point CI, which lets users access the application service. To populate the application service, you then manually add CIs to the new application service.

Before you begin

Role required: app_service_admin

About this task

Manual is one of several methods for populating an application service with CIs. Choosing a method for populating an application service, is only one step of the generic procedure for creating an application service. Ensure that you have completed the initial steps as described in Create an application service. The procedure described here is incomplete by itself as it complements that generic procedure.

For information about the different types of application services and the different methods you can use to populate application services, including Manual, see Application services.

Procedure

1. In the Choose a Method page, select **Manual** as the **Service Population Method**.
2. Fill out the fields that appear, which are specific to the Manual service population method.

Field	Description
Service Population Method	Manual
CMDB Table	Notes the Mapped Application Service [cmdb_ci_service_discovered] table, in which application services, created by the Manual service population method, are stored.
Class	The class from which to choose the entry point **CI** for the application service.

Field	Description
CI	The CI from the specified **Class**, to be the entry point for the application service. **Note:** To eliminate the possibility of delayed results when searching for a specific CI, make your search as specific as possible. A search with *<name> might take a long time and return a large data set.

3. Click **Save**.

What to do next

1. Complete the generic procedure Create an application service.
2. Manually add CIs to populate the application service.

Use the Dynamic Service method to populate application services

The Dynamic Service method for populating an application service generates a dynamic application service. A dynamic application service automatically updates to reflect any changes to CI relationships in the CMDB CI Relationship [cmdb_rel_ci] table.

Before you begin

The Dynamic Service is one of several methods for populating an application service with CIs. Choosing a method for populating an application service, is only one step of the generic procedure for creating an application service. Ensure that you have completed the initial steps as described in Create an application service. The procedure described here is incomplete by itself as it complements that generic procedure.

New dynamic application services initially don't contain any CIs (unless they were converted from legacy business services or legacy manual services). Dynamic application services are automatically populated when CIs that are contained in the service, get connected to other CIs by CMDB relationships. Entry points are automatically created when a relationship between a dynamic application service CI and other CIs is created.

For information about the different types of application services and the different methods you can use to populate application services, including Dynamic Service, see Application services.

Role required: app_service_admin

Procedure

1. In the Choose a Method page, select **Dynamic Service** as the **Service Population Method**.
2. Fill out the fields that appear, which are specific to the Dynamic Service service population method.

Field	Description
Service Population Method	The method used for populating the application service with CIs. Set to **Dynamic Service**.
CMDB Table	Notes the Calculated Application Services [cmdb_ci_service_calculated] table, in which application services created by the Dynamic Service method, are stored.
Levels	The number of levels of related CIs to include in the application service.

3. Click **Save**.

What to do next

Complete the generic procedure Create an application service.

Use Application Services dashboard to monitor health

Use insights from the Application Services dashboard to reduce the number of incomplete application services by populating them and adding missing data. To use application services effectively, ensure that application services are fully configured and are populated.

Before you begin

Role required: itil_admin or app_service_admin

About this task

The dashboard queries for Application Services by checking for those records in the [cmdb_ci_service_auto] class in which the value of Service classification is **Application Service**. Reduce the number of incomplete application services by editing application services and populating any empty attributes. For example, if an application service isn't configured with a service population method, then configure a service population method for it.

The Application Services dashboard is fully integrated into the Insights view in CMDB Workspace and refreshes on a 24-hour cycle during night hours.

Procedure

1. Follow either navigation step to access the Application Services dashboard:
    - Navigate to **Workspaces > CMDB Workspace** and then select **Insights** on the CMDB Workspace menu bar. On the Insights view, select the Application services tile.
    - Navigate to **All > CSDM > Application Service Dashboard**.
2. View the tiles on the Overview tab, which show application services in which basic configurations are incomplete.
    - Total Application Services: Count of all application services, complete and incomplete.

- Population method defined: Count of application services for which a service population method is specified.
- Population method not defined: Count of application services missing a service population method. Data for the tags and top down discovery methods appear only if Service Mapping is installed.

  Note: Some population methods for application services are available only with Service Mapping and therefore won't appear in the tile if Service Mapping isn't installed.

- Application services types: Chart of application services by population methods such as Dynamic CI Group and Manual, including application services without a population method (Empty). The chart includes business services that were converted to application services.
- Application services missing data: Chart of application services by key data that is missing, such as service offering and owner. For example, the bar 'No Service Offering' shows application services without any relationship with a business service or a technical service.

3. View counts on the Application service coverage tab, which shows application services in which other configurations are potentially incomplete.

   - Application Servers: Total number of application servers. The number of those which aren't in any application service and a breakdown of those application servers by class.
   - Databases: Total number of databases. The number of those which aren't in any application service and a breakdown of those databases by class.
   - Hardware Servers: Total number of hardware servers. The number of those which aren't in any application service and a breakdown of those hardware servers by class.

Note: Application servers, hardware servers, and databases, not included in application services, are counted only up to about 100,000, even if the actual count is greater than this limit. This number limit is determined by the value of the glide.cmdb.csdm.app_service.max_results property.

4. Select **Owned by**, **Support group**, or **Change Group** to filter the list of application services that are included in the dashboard, by a key attribute.
   A filter doesn't impact counts of application services in which the respective filter attribute is missing. For example, filtering by Owner, doesn't change the count of the Missing Owner card.
5. Select a tile in either tab on the dashboard to drill down to the associated CIs list view.
   For example, select the Population method not defined tile to see the CIs list view of application services without a population method.

What to do next

Update application services with any missing important details:

1. Navigate to **CSDM > Manage Technology Management Services > Application Services**.
2. In the Application Services list view, select an application service to edit.
3. Add the missing details.

   For details about configuring an application service, see Create application service.

Modify the attributes and relationships required for application services

Modify the lists of attributes and relationships that are required when creating application services.

Before you begin

Role required: itil_admin

About this task

In application service settings, the lists of required attributes and required relationships determine which of those items are required when creating application services. By default, the list of required attributes contains the required Name and Number attributes. Also by default, no relationships are required.

You can choose from predefined lists of allowed required attributes and relationships. To change the requirement status of an attribute, remove it from or add it to the list of required attributes. You can also add Business Application, Technical Service Offering, or Business Service Offering to the list of required relationships.

Procedure

1. Navigate to **All > CSDM > Service Instance Settings**.
2. Review the list of **Available** items in the Required attributes and Required relationships lists and then add items to the **Selected** list.
3. Click **Save**.

Result

Next time that you create an application service, the required attributes and relationships are visibly noted in the Basic Details section on the Create an Application Service page.

Convert business services to application services

Unify the way you manage services in the organization by converting manually created records in the Service [cmdb_ci_service] table into application services. Conversion lets you streamline the different types of services in your organization, leverage ITOM Visibility capabilities, and align with the Common Service Data Model (CSDM). The conversion is irreversible: You can't transform application services back into business services.

Using application services has benefits such as:

- Viewing service maps and change history of services.

- Easily seeing the service context by providing a flat list of all CIs in the application service.
- Monitoring service health. If Event Management is deployed, you can monitor service performance and identify health issues for application services.
- In Change Management, the list of impacted services on a change request form is more accurate because the list includes only application services.
- Applying Customer Service Management tools to open and manage cases at the service level.

Discovery doesn't run on converted application services, because converted services are manual. However, if after the conversion you add **Discoverable by Service Mapping** entry points to the application service, then Service Mapping starts discovering such this application service.

Choosing between application services of the manually created and dynamic type

You can convert business services into application services of the manually created type or of the dynamic type. You can edit manually created application services by adding or removing CIs at any time. The system does not update manually created services automatically. If there are changes to CIs making up a manually created application service, the service does not automatically reflect it.

Dynamic services are updated automatically to reflect any change to CI relationships stored in the CMDB CI Relationship [cmdb_rel_ci] table. When you add a relationship to a CI that is contained in a dynamic service, then that service automatically updates to reflect the addition of the relationship and the associated new CI. In a similar manner, a dynamic service automatically updates upon the removal of a relationship and its associated CI from a CI within the service.

To learn more about different types of application services, see application services.

Conversion process

During conversion, the following changes and processes occur:

- The service record is moved from the Service [cmdb_ci_service] table into the Mapped Application Service [cmdb_ci_service_discovered] table by changing the record class.
- The service instance is set with all the original business service attributes such as name, owner, and operational status.
- The system adds related items from the business service to the converted service instance, up to the specified level.
- The system queries the CMDB for the latest CI changes.
- Event Management, if activated, applies CI impact rules to CIs that are associated with alerts and that are part of the service instance. Event Management deploys CI impact rules for alert monitoring.
- You can edit a converted application service of the manually created type by navigating to **CSDM > Manage Technology Management Services > Application Service**. Then select a converted application service. The service population method for a converted application service, is set to **Converted Business Service**. For more information about editing application services, see Create an application service.

  Note: You can't edit a dynamic application service by adding or removing CIs from it. The system automatically modifies an application service of the dynamic type when you modify relevant relationships for CIs that are part of that application service.

Non-compliant CIs

A conversion might involve adding CIs of the following CI types, which cannot be added to an service instance:

- NAT [cmdb_ci_translation_rule]
- Endpoint [cmdb_ci_endpoint]
- Qualifier [cmdb_ci_qualifier]
- Application cluster [cmdb_ci_application_cluster]

If the original business service contains related items belonging to these CI types, then the system does not add such CIs or connections coming from them. There are system records in the Manual CI Exclusions/

Inclusions [svc_manual_ci_exclusions_inclusions] table. See Components installed with application services for more information.

> **Note:** The Manual CI Exclusions/Inclusions [svc_manual_ci_exclusions_inclusions] table doesn't include CIs added using traversal rules.

Domain separation

In environments with domain separation, only CIs belonging to the same domain as the service instance are added to the service instance. If there is a domain hierarchy, CIs must belong to the same child domain as the service instance.

Convert business services to application services in bulk

Convert a subset of business services to application services, in bulk and automatically rather than one at a time. Individually select the business services for the conversion, and then convert them into application services.

Before you begin

Role required: app_service_admin, ecmdb_admin, or itil_admin

About this task

Use bulk conversion to convert legacy business services to application services. For the bulk conversion, individually select the business services from the Services list view, typically up to 100 business services in a single conversion. You can create multiple bulk conversion records, each with a different set of business services. However, do not include a business service in more than one bulk conversion.

> **Note:** You can't undo this conversion operation.

Procedure

1. Navigate to **All > Configuration > Services**.
2. In the Services list view, select the services that you want to include in the conversion.

3. In the Actions on selected rows drop down list, select **Bulk Convert Application Services**.

4. Fill out the Bulk Convert Services form.

Field	Description
Name	Pre-populated with "Application Service Conversion: <time stamp>".
Select configuration items	List of services included in the conversion. You can unlock the list and select services to remove from the list. Or, use the Select target record search box to search and add services to the list, by service names.
Levels	The number of levels of related CIs to include in the converted application service.
Update service when CMDB updates	Select this check box to convert the business service into an application service of the dynamic type.

5. Select **Start Conversion**.

What to do next

- Check the status of a conversion: On the Bulk Convert Services form, scroll to the Bulk Convert Services Entries section to see the status (such as Ready or Completed) of a conversion.
- Track the progress of a conversion as it runs: In the navigation filter, enter `cmdb_convert_bulk_services.list` and press the Enter key to see the list of conversions, and their progress.

- (Optional) On a change request form, view affected dynamic services. For example, after you add an affected CI that is associated with a dynamic service:

  1. Navigate to **Change > Open**.
  2. Select a new change request to add affected CIs to.
  3. On the Change Request form, scroll to the Related Links section.
  4. Click the **Affected CIs** tab and then click **Add** to add an affected CI to the change request.
  5. Open the form context menu and select **Refresh Impacted Services**.
  6. Click the **Impacted Services/CIs** tab to see any dynamic services that are associated with the affected CI and that are impacted by the change request.

  For more information about affected CIs on a change request, see Associated CIs on a change request.

Convert an individual business service to an application service

Manually convert a specific business service to an application service.

Before you begin

Review the original business service to evaluate it.

- Make sure that all CIs and CI relations are relevant for the future application service. If necessary, change the CI relations in the CMDB.
- Make sure that the original business service doesn't contain more than 5000 CI relations. Application services that contain more than 5000 CI relations cause mapping and monitoring performance issues.
- Decide how many levels of CI relations you are going to use during conversion.

> **Warning:** The conversion is irreversible: You can't transform application services back into business services.

Role required: app_service_admin

Procedure

1. Navigate to **All > Configuration > Business Services**.
2. Select the business service that you want to convert to an service instance.
3. Click **Convert to Application Service**.
   The Converting to Application Service dialog box opens.
4. Select a number from the **Up to** list, as the number of levels of related CIs to include in the conversion.
   The maximum number of levels is eight.
5. Select **Update service when CMDB updates** to convert the business service into an application service of the dynamic type.
6. Click **OK**.

Result

The system adds the CIs from the business service to the converted service instance.

What to do next

Open the map for the newly converted service instances.

Make sure that the service instances aren't too large:

- Service Mapping doesn't offer to view CI list instead of a map for a service instance.
- There is no discovery message indicating that the service instance is too large: The map does not display the entire service, because it is too large. The number of CI connections exceeded the allowed maximum.

If the service is too large, perform the following actions:

- Review the converted service instance to identify CI relations irrelevant or redundant for this service. Remove such CI relations in the CMDB.
- Decide how many levels of related CIs you must include into this service instance. If necessary, change the number of levels used in conversion to reduce the service size.

Related concepts

- Application services

Convert legacy manual services to dynamic application services

Unify the way that you manage services in your organization by converting legacy manual services into dynamic application services. Conversion lets you streamline the different types of services in your organization, leverage ITOM capabilities, and align with the Common Service Data Model (CSDM).

Before you begin

Role required: app_service_admin

About this task

Note: Converting a legacy manual service to a dynamic application service is irreversible. Once converted, you can't revert it back to a legacy manual service. However, you can convert a dynamic application service to a manual application service.

You can't edit a dynamic application service directly. If you remove a CI from the map, it reappears once the service is recalculated. To ensure permanent changes, update the CI relationships in the CMDB CI Relationship [cmdb_rel_ci] table.

During conversion, the following changes and processes occur:

- A change to the record class moves the service record from the Service [cmdb_ci_service] or the Manual Service [cmdb_ci_service_manual] table to the [cmdb_ci_service_calculated] table.
- The dynamic application service is configured with all the original attributes of the legacy manual service such as name, owner, and operational status.

- Related items from the legacy manual service are added to the converted dynamic application service, up to three levels by default.
- All connections created between CIs in the dynamic application service are endpoint CIs with the relationship uses, implement, or application flow.
- Event Management, if activated, applies CI impact rules to CIs that are associated with alerts and that are part of the application service. Event Management deploys CI impact rules for alert monitoring.

A conversion might involve adding non-compliant CIs, which can't be added to an application service:

- NAT [cmdb_ci_translation_rule]
- Endpoint [cmdb_ci_endpoint]
- Qualifier [cmdb_ci_qualifier]
- Application cluster [cmdb_ci_application_cluster]

If the original manual service contains related items belonging to these CI types, then these CIs or connections coming from them aren't added to the dynamic service. If necessary, you can prevent other CI types from being added to application services by modifying the Manual CI Inclusions/Exclusions [svc_manual_ci_exclusions_inclusions.list] table.

Procedure

1. Navigate to **All > Configuration > Application Services > Application Services**.
2. Confirm that the view is set to the default view.
    a. Select the List controls icon for the list view.
    b. Select **View** and then select **Default view**.
3. Open the legacy manual service that you want to convert to a dynamic application service.
4. In the Related Links section on the service form, select **Convert to Dynamic Service**.

Manually add CIs to an application service

Add configuration items to manually created application services or to services discovered by Service Mapping.

Before you begin

- Verify that the CI type for the configuration item (CI) that you are planning to add, exists. If necessary, create the CI type as described in Create CI types for Service Mapping and Discovery.
- Add CIs to the CMDB for the device or application that you want to add, if necessary. See Populate the CMDB for more information.

Role required: app_service_admin or service_mapping_admin

About this task

Adding a CI to an application service requires creating a relationship between the new CI and a CI in the application service. You can add CIs to an application service that was created manually, by either:

- Adding a method to populate the application service.

  Navigate to **CSDM > Manage Technology Management Services > Application Service**. Select an application service and then use the **Populate the Application Service** tab to choose a method to populate the application service. For more details, see Create an application service.

- Using the application service service map as described in the steps below.

  The default relationship type of the added connection in this case is Depends on::Used by. You can modify this default relationship type by changing the value of the sa.it_service.manual_ci_rel_type property. See Components installed with application services for more information.

Important: You cannot fine-tune or edit tag-based and dynamic services from the map.

Manually adding a new CI to an existing CI in a service instance prompts the CMDB to update the information about both CIs, including their relationship type. If other application services use the same applicative flow, the CMDB recognizes the new CI and automatically adds it to those services as well.

For example, you manually add an IBM WebSphere Message Broker to an IBM WebSphere HTTP Listener in the Bank Customer Portal service. The system also adds this IBM WebSphere Message Broker to the same HTTP Listener in the Bank Internal Portal service, because it uses this HTTP Listener. Similarly, when you remove a CI you added manually, the system removes it from all application services where you either manually added it or it was automatically added by analogy.

You can manually connect a CI only to actual CIs existing in the CMDB, not to a visualization of other items on the map such as clusters or boundaries. Also, you cannot add CIs of these CI types to an application service:

- NAT [cmdb_ci_translation_rule]
- Endpoint [cmdb_ci_endpoint]
- Qualifier [cmdb_ci_qualifier]
- Application cluster [cmdb_ci_application_cluster]

There are system records in the Manual CI Exclusions/Inclusions [svc_manual_ci_exclusions_inclusions] table. See Components installed with application services for more information.

Note: The Manual CI Exclusions/Inclusions [svc_manual_ci_exclusions_inclusions] table doesn't include CIs added using traversal rules.

In environments with domain separation, only CIs belonging to the same domain as the service instance are added to the service instance. If there is a domain hierarchy, CIs must belong to the same child domain as the service instance.

If working with an service instance discovered by Service Mapping, manually add a CI:

- To indicate that an service instance contains a device or application, which Service Mapping cannot discover. For example, add an A/C unit to the Production Floor service.
- To add a temporary placeholder for a CI, which Service Mapping did not discover. In this case you are planning to perform necessary troubleshooting to ensure that Service Mapping discovers this CI in the future. For example, add an IBM WebSphere Message Broker to the Bank Customer Portal service.
- To create an service instance that combines entry points and CIs automatically discovered by Service Mapping with entry points and CIs from the CMDB. After you manually add an entry point, you can update the service instance with CIs from the CMDB based on the relationships defined there.

For additional information related to Service Mapping, see Pattern customization and Enable traffic-based discovery for CI types or specific CIs.

Procedure

1. Open the service instance map.
    a. Navigate to **All > CSDM > Manage Technology Management Services > Service Instance**.
    b. Select the needed service instance.
    c. On the service instance page, select **View Map**.
2. If needed, click **Edit** to ensure that the map is in Edit mode.

    If Service Mapping is deployed, then in Edit mode, the Discovery Messages section appears below the map.
3. To connect a CI to another CI on the map, right-click the CI to which you want to connect the new CI, and then select **Add a CI**.
4. In the Add A CI dialog box, specify the CI to add:

Field	Description
CI Type	Select the CI type (CI class) for the CI you are adding. Every CI belongs to a CI type which contains a set of attributes configured for this kind of CI, for example, cmdb_ci_appl for applications.
CI Name	Select the CI from the list of CIs of the selected CI type. **Note:** To eliminate the possibility of delayed results when searching for a specific CI, make your search as specific as possible. A search with *<name> might take a long time and return a large data set.

The CI type list includes only allowed CI types. For example, you cannot add an application cluster.

5. Click **Submit**.
   The manually added CI appears on the map.

   Note: When you manually add a CI, which is an application, as a child to a service that already includes its parent application CI, the newly added child application CI is hidden inside the inclusion. Click the plus (+) symbol next to the parent application CI to see the child application CI.

6. (Optional) If Service Mapping is activated, add a discoverable outgoing connection for the manually added CI:

   a. Right-click the manually added CI.

   b. Select **Manually add a connection**.

Note: If you do not see the **Manually add a connection** option in the right-click menu, check that you are logged in with the user that belongs to the same domain as the application service.

c. Configure attributes for the entry point as described in Entry points attributes.

d. Click **Submit**.
Discovery and Service Mapping attempt to discover this CI. If successful, the CI appears on the map. Otherwise, a warning icon (⚠) appears.

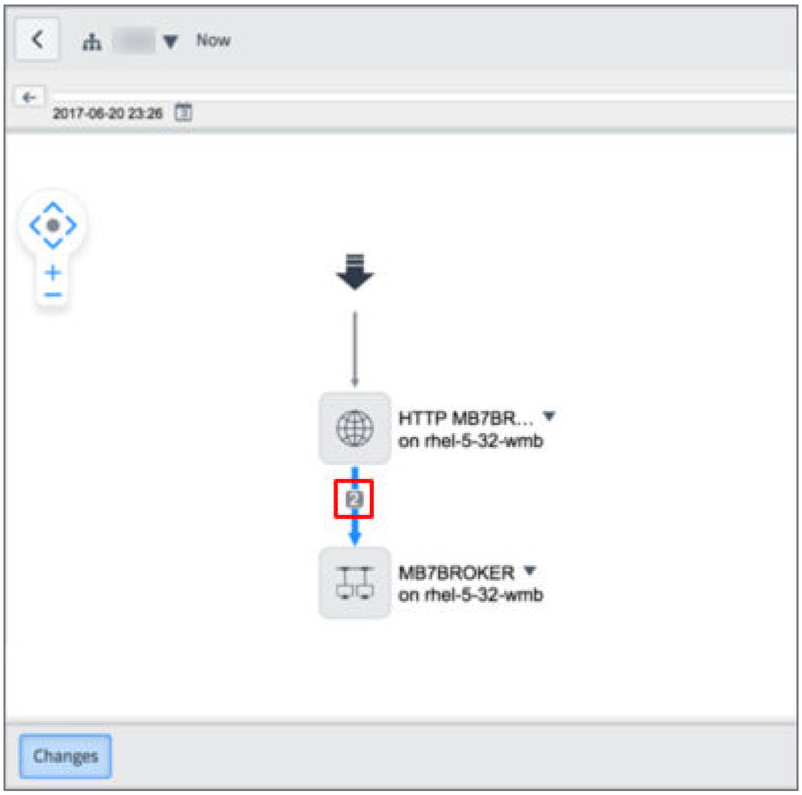
7. (Optional) If Service Mapping is activated and you want Service Mapping to automatically discover a CI, which you previously added manually:

   a. Customize the relevant pattern or fine-tune traffic-based discovery to enable Service Mapping to discover the CI.

   b. Navigate to the relevant service instance map.

   c. Click **Run discovery**.

   d. After the discovery process finishes, verify that Service Mapping discovered the CI by checking the connector leading to the CI. If Service Mapping discovered the CI, then two connectors, a manual and automatically discovered, appear for the CI.



e. Right-click the CI you added manually.
   In the example, it is IBM WebSphere Message Broker.

f. Select **Remove manually added CI**.
   The map shows the CI with only one connector leading to it. If this CI had any manually added connections, they are removed together with the manually added CI.

Related tasks

- Link application services

Related topics

- addCI()

Manually update an application service with changes from the CMDB

Ensure that an service instance is up-to-date and reflects all the latest changes to its configuration items (CIs). Regularly update application services to reflect any changes to CIs and their relationships in the CMDB.

Before you begin

Role required: app_service_admin

About this task

There is no mechanism or an API that automatically updates application services that were created manually. Also, you may need to manually update application services discovered by Service Mapping, if they contain manually added CIs. You can only update application services which contain manually created entry points and which are not discovered by Service Mapping.

An example of a change is deleting CIs from the CMDB or connecting two CIs one of which is part of an application service. In the first case, your application service may show a CI that no longer exists. In the second case, on the contrary, the application service omits a CI.

An update might involve adding CIs of the following CI types, which cannot be added to an application service:

- NAT [cmdb_ci_translation_rule]
- Endpoint [cmdb_ci_endpoint]
- Qualifier [cmdb_ci_qualifier]
- Application cluster [cmdb_ci_application_cluster]

There are system records in the Manual CI Exclusions/Inclusions [svc_manual_ci_exclusions_inclusions] table. See Components installed with application services for more information.

> **Note:** The Manual CI Exclusions/Inclusions [svc_manual_ci_exclusions_inclusions] table doesn't include CIs added using traversal rules.

Also, the system can connect a CI from the service instance only to actual CIs that exist in the CMDB, not a visualization of other items on the map like clusters or boundaries.

The maximum number of CI connections added to application services during this operation is controlled by the sa.service.max_ci_service_population property. By default, the value is 1,000 (one thousand connections). Increasing the number of CI connections may cause performance issues. To adjust the maximum number of added CI connections, add the sa.service.max_ci_service_population property, as described in Add a system property.

In environments with domain separation, only CIs belonging to the same domain as the service instance are added into the service instance. If there is a domain hierarchy, CIs must belong to the same child domain.

You can also update application services by using APIs.

Procedure

1. Navigate to **All > CSDM > Manage Technology Management Services > Application Service**.
2. On the Application Services list view, select the service instance that you want to update.
3. Click **Advanced** and then click **Advanced Configurations**.
4. On the Additional Info page, click the **Update with changes from CMDB** related link.
5. Select a number in the **Up to** list to limit the number of levels of related items to be updated.
   If the specified number is higher than the number of levels of related items that already exist in the service instance, then the system adds the missing CIs and their connections.

   > **Warning:** Specifying a lower number than the number of levels that already exist in the service instance, does not result in the removal of CIs from the service instance.

6. Click **OK**.

Result

- The system updates the service instance with the changes from the CMDB and shows them on the map.
- After the update process is complete, the service instance form opens.

Link application services

You can manually link two application services by adding a reference to one application service into another application service. The service that contains the reference, becomes a dependent service. The service that you include as a reference is a contained service. You can link application services to create dependencies for impact monitoring in Event Management.

Before you begin

You can edit discovered and manually created service instances.

> **Important:** You cannot fine-tune or edit tag-based and dynamic services from the map.

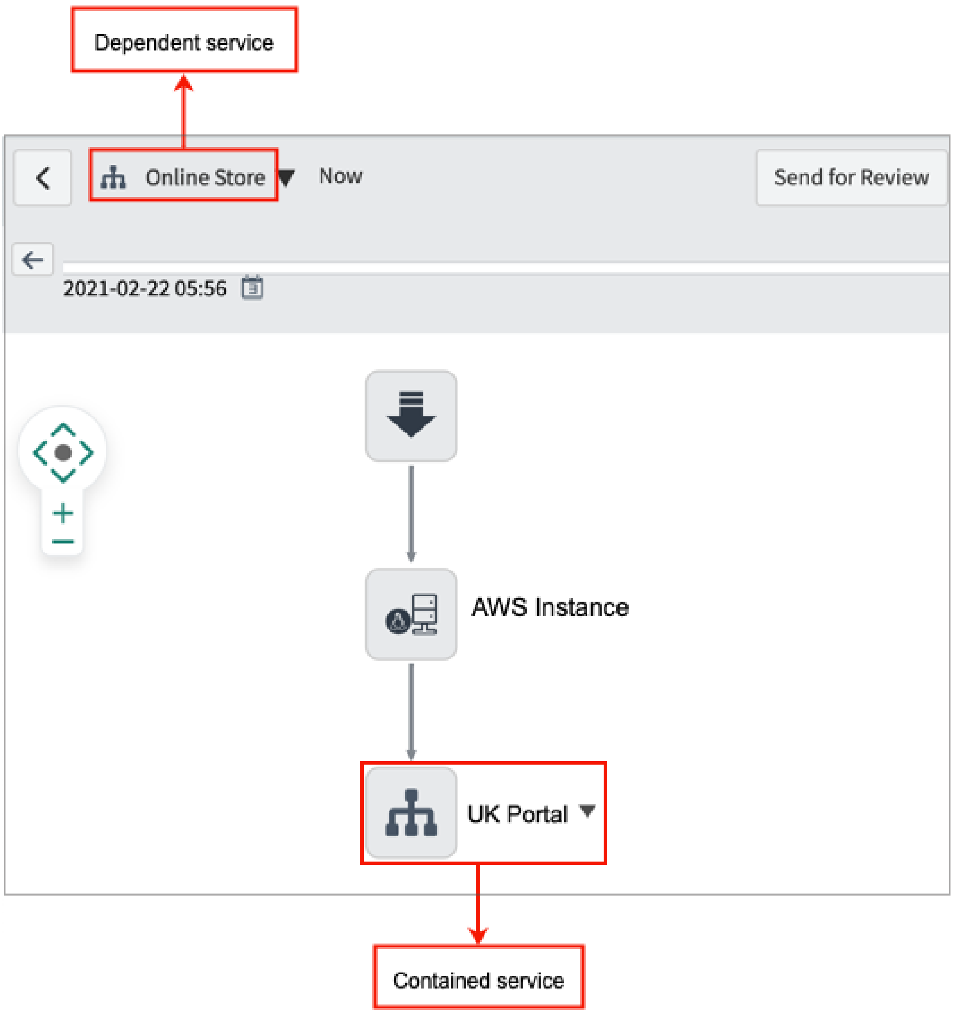
Ensure that you know the name and the service type of the application service, to which you want to add a reference.

Role required: app_service_admin or service_mapping_admin

About this task

To create a link, add a reference to the relevant application service as an outgoing connection of the relevant CI inside another application service. For example, you can add the UK Portal application service as a link to the Online Store application service. In this case, the Online Store service becomes dependent on the UK Portal service that it contains. The Online Store service reflects discovery errors for its contained service in the Edit map mode, as well as alerts in Event Management.

Example of linked application services



When you link an application service to another application service, the information about the CI, to which you linked the service, is updated in the CMDB. The CMDB recognizes other application services that use the same applicative flow, and adds the contained application service to these application services by analogy. The same logic applies when you remove a contained application service: The system removes it from all application services where you either manually linked this service or the system linked this service by analogy.

When using Service Mapping, you may want to link application services to create:

- A dependency between two application services.
- A placeholder for a map branch that Service Mapping failed to discover. If you create or customize a pattern to discover the configuration item (CI) serving as an entry point for the contained service instance, Service Mapping can discover this contained service.
- An indication that an service instance contains a branch, which Service Mapping cannot discover.

You can add an service instance as a contained service to as many service instance as necessary.

Procedure

1. Navigate to **Service Mapping > Application Services**.
2. Click **View map** next to the relevant application service.
3. If needed, click **Edit** to ensure that the map is in Edit mode.
4. Right-click the CI to which you want to link an application service as a reference.
5. Select **Add A CI**.
6. In the Add a CI dialog box, select the application service you want to add as a contained service:

Field	Description
CI Type	Select the relevant service type from this list: • `Tag-Based Application Service` • `Mapped Application Service` for discovered or manually created application services

Field	Description
	• `Calculated Application Service` for dynamic services • `Dynamic CI Group`
CI Name	Select the name of the application service that you want to link as a contained service.

7. Click **Submit**.
   The icon for the contained service appears on the map.

Related concepts

- Application services

Related topics

- View dependent application services in classic Service Mapping
- View contained application services in classic Service Mapping

Group application services

Organize application services by groups to perform actions simultaneously on multiple services, and to control user access to services. You can use Event Management to track service health by service groups.

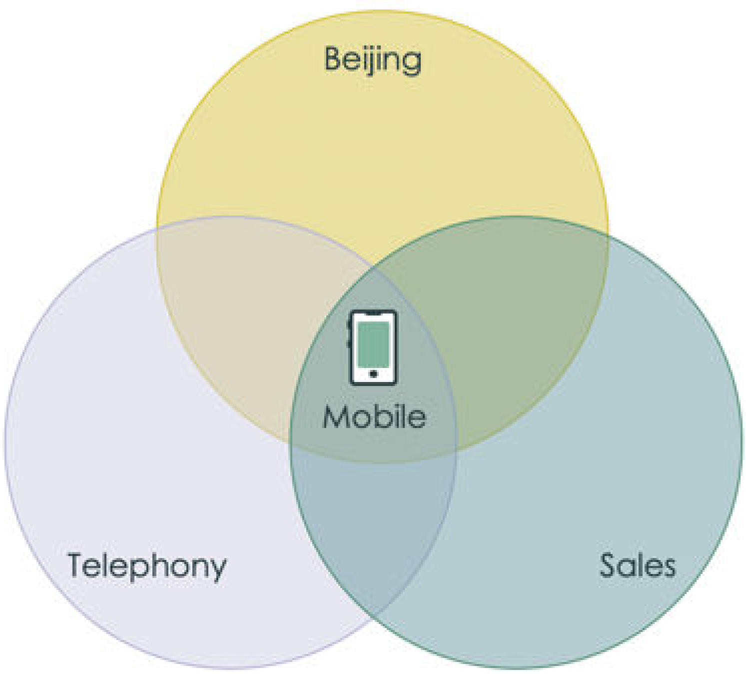
Before you begin

Role required: service_mapping_admin or app_service_admin

About this task

Typically, enterprises have hundreds of services which makes it impractical to manage them individually. Service groups can make service lists much shorter and easier to manage, especially in large organizations or service providers.

How you group service instances depends on the user and on service provisioning policies in your enterprise. The relation between service instances in groups is purely logical and the same service instance can belong to multiple groups. For example, the Mobile service can be part of the following service groups: Sales, Beijing, and Telephony.
Example of a service instance belonging to different groups



You can embed a service group within another service group to create a hierarchy of service groups. If users have access to a parent service group, they automatically have access to all its child groups. By default, all new services are assigned to the **All** service group that lets all users view and manage service instances. When you assign a role to a service group, the users with this role can access service instances in this service group and in the **All** service group. To enable users with this role to access other services, assign this role to the respective service group. Do not assign user roles directly to the **All** service group.

If Service Mapping is activated, service groups can contain a mixture of manually created application services and application services discovered by Service Mapping.

You can use ServiceNow AI Platform Notifications to alert users if the service group severity changes to critical. The overall severity of the group is determined by the highest alert severity within the group.

Procedure

1. Navigate to **All > Configuration > Application Services > Service Groups**.
2. Click **New**.
3. Enter the name of the new service instance group in the **Name** field.
4. To embed this group in another group, enter the name of the other group in the **Parent Group** field.
5. Right-click the form header and click **Save**.
6. Add a service instance to the newly created service group.
    a. In the Service Group Members section, click **New**.
    b. In the **Name** field, enter the name of the service instance.
       If you are using Event Management, you can also enter an alert group name.
    c. Click **Submit**.
7. Alternatively, add a service instance to a group from the service instance form.
    a. Navigate to **All > Configuration > Service Instances > Service Instances**.
    b. Select the service instance you want to add to a service group.
    c. In the **Service Group Members** section, double-click **Insert a new row**.
    d. Enter the name of the service group to which you want to add the selected service instance.

e. Click the **OK** icon ( ).

f. Click **Update**.

Control user access to application services

Assign user roles to service groups to grant users access to application services in your organization. Your organization may restrict access to some services for security or secrecy reasons.

Before you begin

Make sure that you have performed the user provisioning tasks for the users you want to grant access:

1. Add users to user groups.
2. Create new roles.
3. Assign roles to users or user groups.

Also, make sure that you have created service groups as described in Group application services.

Role required: app_service_admin or service_mapping_admin

About this task

In the base system, the following roles provide access to application services:

app_service_admin

Creates and modifies service instances, creates service groups, views, and edits service instance maps.

app_service_user

Views maps for operational service instances and retrieves service content using the getContent - GET REST API. The itil role that serves as the basic helpdesk technician role contains the app_service_user role.

Service Mapping provides these preconfigured roles:

service_mapping_admin

Sets up the Service Mapping application. Maps, fixes, and maintains service instances. Also performs advanced configuration and customization of the product. Assign this role to application administrators.

service_mapping_user

Views maps for operational service instances to plan change or migration, as well as analyze the continuity and availability of services. Assign this role to application users.

sm_app_owner

Provides information necessary for successful mapping of a service instance. Once a service is mapped, this user reviews the results and either approves it or suggests changes. Assign the sm_app_owner role to users who own service instances and are familiar with the infrastructure and applications that make up the services.

Note: Users with the itil role only can view all service instances.

Event Management provides these preconfigured roles:

evt_mgmt_admin

Has read and write access to all Event Management features to configure Event Management.

evt_mgmt_operator

In addition to the evt_mgmt_user permissions, can also activate operations on alerts such as acknowledge, close, open incident, and run remediations.

evt_mgmt_user

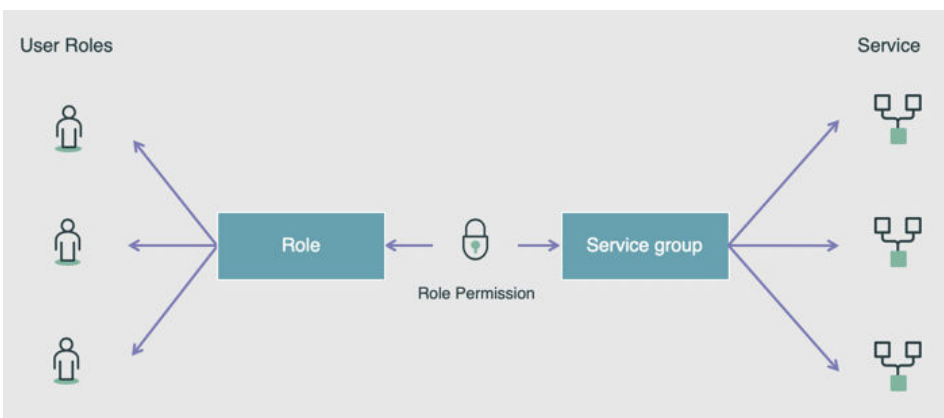
Has read access to all Event Management features. Has write access to alerts to manage the alert life. Has the itil role to be able to manage incidents that are created from alerts.

evt_mgmt_integration

Has create access to the Event [em_event] and Registered Nodes [em_registered_nodes] tables to integrate with external event sources.

Typically, enterprises have hundreds of services which makes it impractical to manage them individually. Service groups can make service lists much shorter and easier to manage, especially in large organizations or service providers. In a hierarchy of service groups, access to a parent service group automatically grants access to all the child service groups.

Users inherit permissions from roles that are assigned to them. You can assign some roles directly to service groups to allow all users with this role to access all application services belonging to this group. However, most enterprises choose to organize their roles as a hierarchy. It helps to manage roles across multiple ServiceNow applications. For example, the Service Mapping administrator [service_mapping_admin] can be part of a broader administrator role like administrator [admin]. You can add users to user groups and then assign roles to the user groups to give permissions of this role simultaneously to all the group users.
Assigning a role to an application service group



By default, all new services are assigned to the **All** service group that lets all users view and manage service instances. When you assign a role to a service group, the users with this role can access service instances in this service group and in the **All** service group. To enable users with this role to access other services, assign this role to the respective service group. Do not assign user roles directly to the **All** service group.

Procedure

1. Navigate to either of the following:
   - **Configuration > Application Services > Service Group Responsibilities**.
   - If Service Mapping is activated: **Service Mapping > Services > Service Group Responsibilities**.
   - If Event Management is activated: **Event Management > Services > Service Group Responsibilities**.
2. Click **New** and fill out the Application Service Group Responsibilities form.

Field	Description
Application Service Group	Service group to which you want to assign a role.
Role	Role you want to assign to the selected service group. For example, financial_services_admin.

3. Click **Submit**.

Example

To manage access to services that contain sensitive financial information in your organization:

1. Organize the services into the Financial Services group.
2. Create a new user role, financial services administrator [financial_services_admin] role, that contains the [app_service_admin] role.

3. Assign the Financial Services administrator role to the Financial Services group.

As a result, only users with the Financial Services administrator role can access application services belonging to the Financial Services group.

View an application service map in base system

An application service map provides a visualization of data for the CIs comprising an application service, and the relationships and connections between these CIs.

Before you begin

Role required: app_service_user to view the map in View mode, and app_service_admin to modify services in Edit mode.

About this task

When you create an application service, the system generates an associated application service map. The system then updates the map to reflect any changes to the application service. This map consists of icons representing CIs and arrows that represent the connections between them.

If Service Mapping is deployed, see Application service maps and View CI connection attributes in an application service map for more details.

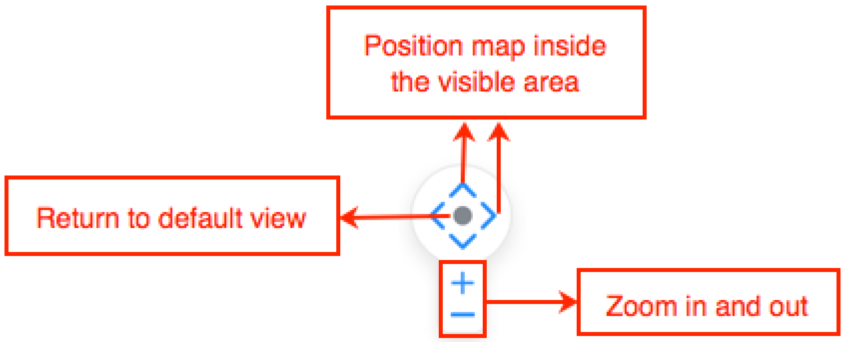
To open an application service map, navigate to **CSDM > Manage Technology Management Services > Application Service**, select an application service, and then click **View map**.

Perform any of the following operations in the application service map.

Procedure

- Select ▼ on the windows bar to navigate to a different application service.
- Use the navigation tools to increase or decrease the view of the map and to move the map on the page.

You can also click anywhere on the map area and drag a segment of the map into the visible area.



- View changes: You can view changes and change records associated with the application service as a whole or with any of its CIs, within a time range.

  For more information, see View the change history of application services.

  Records under the **Change** tab underneath the map, which are associated with a selected CI or connection, are highlighted. If you select a change record under the **Change** tab, then the associated CI icon appears yellow on the map.

- View attributes: When you select a device, application, or connector on the map, it appears in blue and its attributes appear in the Properties pane on the right of the map.

  When nothing is selected on the map, the details of the application service itself appear on the Properties pane.

  Open the CI's form for further details by clicking **Detailed Properties** at the bottom of the Properties pane.

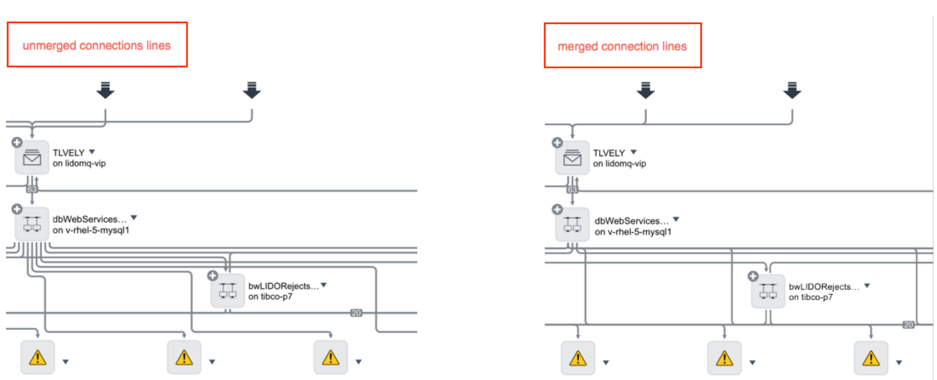
- Select **Edit** to add or remove CIs from the map or **View** to switch the map mode.

- Select  for **Additional actions**:

- Set **Group CIs on map**: Simplify maps by grouping 10 or more CIs belonging to the same type and hosted on servers sharing prefix and domain name.
- Set **Spanning tree view**: Simplify the map by organizing CIs into a tree structure and hiding some connection lines. This option is especially useful for very large maps.
- **Map Indicators**: Show additional information for a CI or for the application service itself by displaying related records such as alerts, outages, incidents, and problems. For each indicator that is enabled, the corresponding indicator icon appears next to CIs with associated records, and the corresponding tab appears underneath the map. If a record is associated with the application service itself, the indicator appears next to the application service name.

  For information about managing map indicators, see Create or modify map indicators. For more general information, see Event Management Map Indicators (Video).
- **Export to PDF**: Export the map to a .PDF file which you can then share as needed. After the PDF file is ready, click to download the PDF file to your local drive.

- View the details of a connection.
  By default, connection lines for the same CI on an application service map, are merged. This merge reduces clutter on the map and helps to make the map more readable. For a merged connection line, you can view details for all the underlying connection lines.
  Merged connection lines



- To view the source and target CIs of a connection, right-click a connection line.
  If spanning tree view is enabled:
  1. Select the CI whose connections you want to view to show all the concealed connections for the CI.
  2. Right-click one of the connection lines.
- To view properties of a connection, click a connection line. For manually added connections, Endpoint Type is Manual Endpoint.
- To view properties of a connection within a merged connection:
  1. Right-click the merged connection line.
  2. Select one of the connections.
  3. Select **Select edge**.

What to do next

You can change the details that appear in the Properties pane by updating the form view 'Form view and section', as described in Configuring the form layout.

Related reference

- Spanning tree view property

View CI attributes in an application service map in classic Service Mapping

An application service map displays attributes for each configuration item (CI) that is part of the application service, as well as for the application service itself. The attributes come from the CMDB.

Before you begin

Role required: service_mapping_admin or service_mapping_user

About this task

You can view the following information for each CI:

Name label

The CI name. This attribute is either preconfigured on the CI or configured during CI installation.

Basic attributes

A summary of the most important CI attributes.

Detailed attributes

A complete list of all attributes collected for the CI.

Each CI type (CI class) has different attributes. For example, the Linux Server type has different attributes than the SQL Instance type.

If Service Mapping is deployed, the way CIs appear on the map depends on the view you select for the map. Attributes available for viewing also depend on the Service Mapping setup. For more information, see description of components installed with Service Mapping.

Procedure

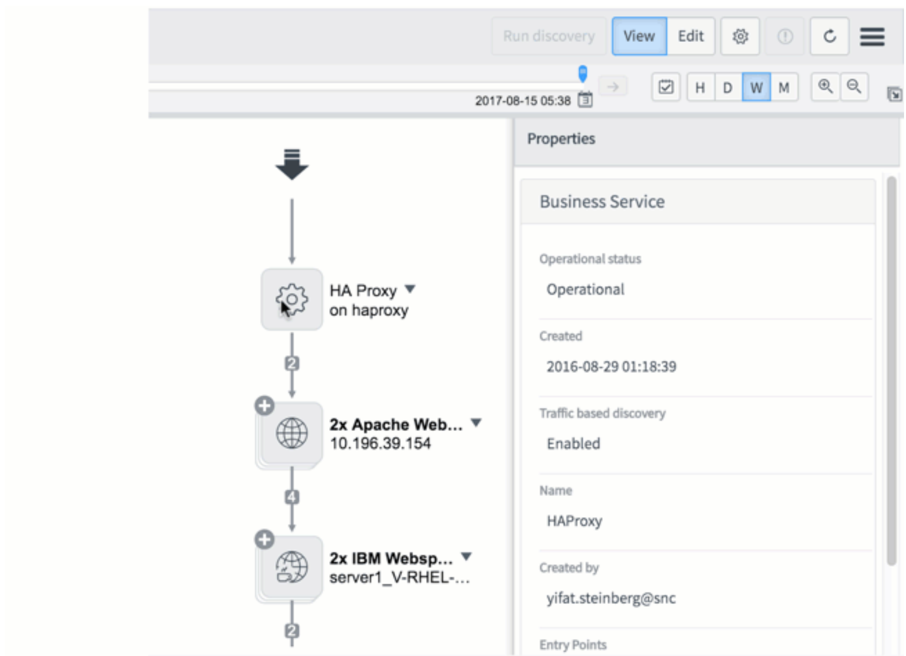
1. Open the service instance map.
    a. Navigate to **All > CSDM > Manage Technology Management Services > Service Instance**.
    b. Select the needed service instance.
    c. On the service instance page, select **View Map**.
2. If needed, click **Edit** to ensure that the map is in Edit mode.

    If Service Mapping is deployed, then in Edit mode, the Discovery Messages section appears below the map.
3. To see the full name of a CI whose name has been shortened on the map, point to the CI.

A tooltip displays the full CI name.

4. Click a CI to see its details in the **Properties** pane.

   The attributes of applications and the servers that host them appear separately.



5. To view more detailed attributes for the CI, click **Detailed properties** at the bottom of the Properties pane.

6. (Optional) To view configuration files associated with a CI in environments where Service Mapping or Discovery are enabled and tracking changes to CI configuration files is enabled):

   - Review the list of files under **Tracked Files** in the **Properties** pane. Click the file name to open the actual file.
   - Click the **Affected CIs** tab and view the list of configuration files. Click the file name to open the actual file.

View the change history of application services in classic Service Mapping

You can view the changes made to an application service as a whole and to the individual configuration items (CIs) comprising the service. Change history is useful for maintenance, planning, or troubleshooting procedures.

Before you begin

Role required: admin, service_mapping_admin, service_mapping_user, app_service_admin, or app_service_user

About this task

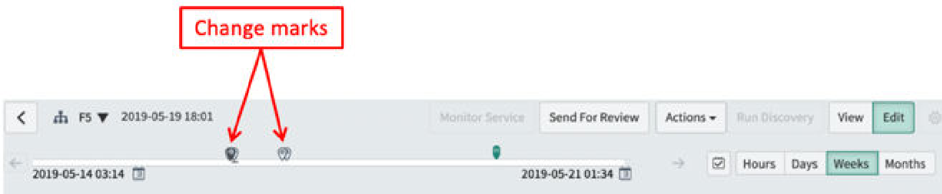
Details about changes to a service instance and to its CIs are stored in the CMDB. Typically, these changes reflect adding or removing CIs from a service instance, upgrading or updating CIs, or modifying CI configuration files. The system gathers this data by querying CMDB tables and then creating the change history view. In deployments where Service Mapping is activated, the type of change information Service Mapping queries depends on discovery patterns that Service Mapping uses to discover CIs.

Changes to configuration files are associated with CIs to which these files belong. Maps show configuration file changes as changes to related CIs.

While you can see change records for a specific CI in the context of application services, you can also see detailed history of a specific CI separate from its service instance as described in History Timeline.

If the ServiceNow AI Platform is configured to validate changes, all changes are evaluated and rendered as valid or not. If a change is valid, its change record on the service instance map is marked as approved. For more information about configuring the platform for change validation, see Managing proposed changes.

Changes to the service instance appear on the history timeline.



The type of change mark depends on the nature of changes that it represents:

Light gray balloon (**)**

Unapproved change that does not influence the service instance behavior. For example, a change in a network path or adding a node to a cluster.

Dark gray balloon (**)**

Unapproved change that changes the service instance behavior.

Green balloon (**)**

An approved change in deployments where the ServiceNow AI Platform is configured to validate changes.

Double balloon (**)**

Multiple separate changes that happened a short time from each other.

You can mark times on the history scale by creating baselines to quickly return to the marked view.

Procedure

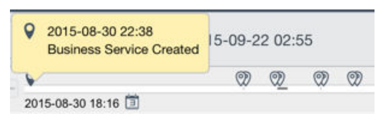
1. Open the service instance map.
   a. Navigate to **All > CSDM > Manage Technology Management Services > Service Instance**.
   b. Select the needed service instance.
   c. On the service instance page, select **View Map**.

2. Review change records created for this service instance on the **Changes** tab at the bottom of the page.

   If Service Mapping is deployed, then in Edit mode, the Discovery Messages section appears below the map.

3. On the history timeline, set the time range of changes that you want to view.

Option	Action
To set the time range of the history timeline	Click the hour, day, week, or month icons. 
To increase or decrease the time range	Click the zoom in and zoom out icons. 
To change the upper limit on your history range	Click the history scale.  The time that serves as the upper limit appears above the history timeline.

Option	Action
	Note: You cannot set the lower limit on your history range to a time before this service instance was created. This time is marked with the **IT Service Created** event on the history timeline. 

The map shows the history view of the service instance for the time you selected.

> **Note:** The **Change** tab shows all change records, even the ones which are filtered out of the history view.

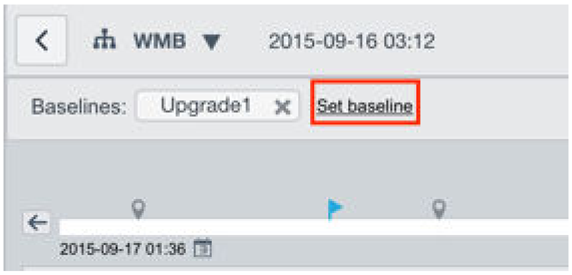
4. To mark a time on the time scale, set a baseline:

   a. Click the **Compare** icon.

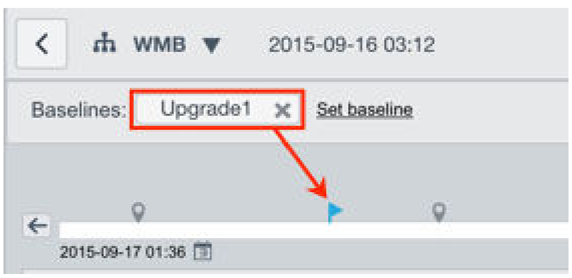
   

   b. Navigate to the time you want to mark as a baseline on the history scale.

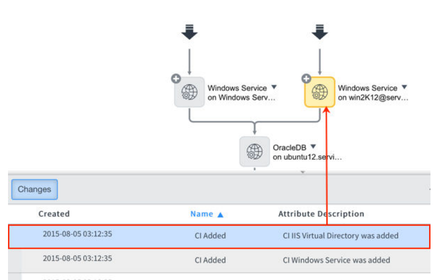
   c. Click **Set baseline**.



d. Enter the name of the baseline and click **OK**.
   The new baseline appears as a button above the history scale and as a blue flag on the history scale.

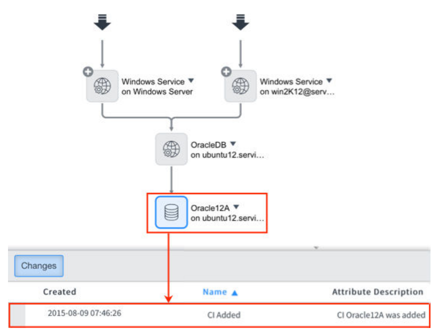


5. View the change history.

Option	Action
To see the CI responsible for a change record	Select a change record on the **Changes** tab. The related CI is marked yellow in the map. 

<table>
<tr><th>Option</th><th>Action</th></tr>
<tr><td>To see only change records related to a CI</td><td>Select the required CI or the connection on the map.

The Changes tab displays only change records related to the selected CI or connection.


</td></tr>
<tr><td>To see the configuration file at the selected moment in the past</td><td>a. Set the time on the history scale.

b. In the Properties pane, scroll to Tracked Configuration Files, and click the file name.

The new tab opens displaying the content of the tracked configuration file at the selected time.</td></tr>
<tr><td>To see the network at the selected moment in the past</td><td>a. Set the time on the history scale.

b. Right-click the connection and select Show network path.</td></tr>
</table>

Option	Action
	The new tab opens displaying the network or storage path map for the time you selected. **Note:** You cannot view the network path for connections marked as boundaries to this service instance.

6. To exit the history view and see the current status of the service instance, click the current icon.

Related tasks

- View an application service map in base system
- Compare two versions of an application service in classic Service Mapping

Related topics

- Modify tracking changes in configuration files
- Fine-tune tracking changes for the change history

Compare two versions of an application service in classic Service Mapping

You can see a summary of application service changes at a glance by comparing two versions of an application service. This feature is useful for checking the application service status before and after a certain change or problem.

Before you begin

Role required: admin, app_service_admin, app_service_user, service_mapping_admin, or service_mapping_user

About this task

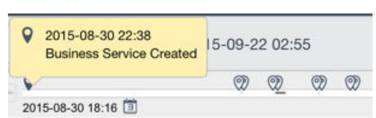
Specify two points in time for which to compare the two versions of an application service. You can use the change indicators on the timeline to specify one point in time that is before and another that is after a change for which to see the details. For example, if you know that the application service started to fail at a certain time, you can compare two versions of the application service, one before and one after the problem started. This comparison lets you see the summary of changes that possibly led to the problems.

Service Mapping, if deployed, tracks and shows all changes to a CI including configuration files associated with a CI. When you compare two versions of an application service, you can see changes made to configuration files as changes to CIs. You can also compare two versions of a configuration file to see the actual changes in the files, during the time range specified for the comparison.

Procedure

1. Open the service instance map.
    a. Navigate to **All > CSDM > Manage Technology Management Services > Service Instance**.
    b. Select the needed service instance.
    c. On the service instance page, select **View Map**.
2. If needed, click **Edit** to ensure that the map is in Edit mode.

    If Service Mapping is deployed, then in Edit mode, the Discovery Messages section appears below the map.
3. On the history timeline, set the time range of changes that you want to view.

Option	Action
To set the time range of the history timeline	Click the hour, day, week, or month icons. 
To increase or decrease the time range	Click the zoom in and zoom out icons. 
To change the upper limit on your history range	Click the history scale.  The time that serves as the upper limit appears above the history timeline. **Note:** You cannot set the lower limit on your history range to a time before this service instance was created. This time is marked with the **IT Service Created** event on the history timeline. 

The map shows the history view of the service instance for the time you selected.

Note: The **Change** tab shows all change records, even the ones which are filtered out of the history view.

4. Click the **Compare** icon.

5. Set **Compare point 1** and **Compare point 2** as the two points in time for the comparison.

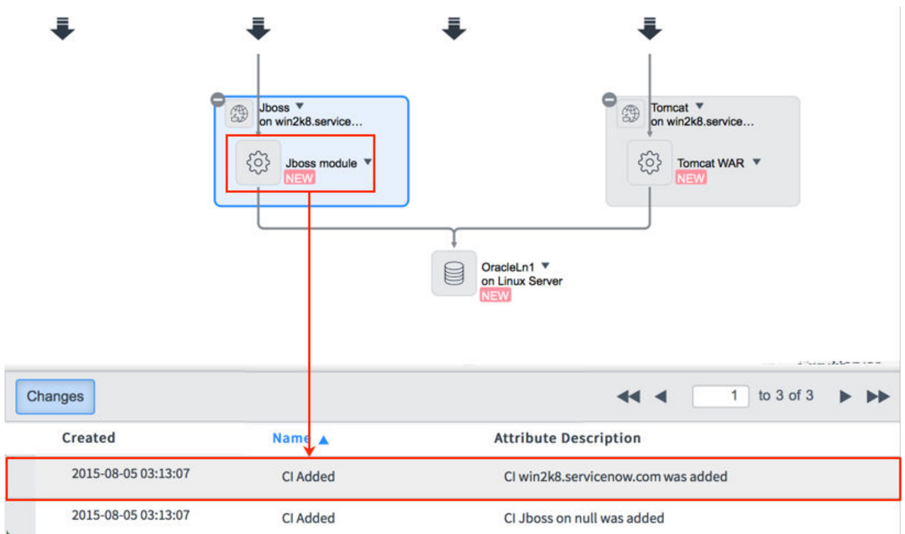
   You can drag the pointers on the history scale to set corresponding time points.

   If the history scale does not include the time set for comparison, then its corresponding pointer appears next to the compare point in yellow:

   Note: If there are no changes to the service during the time interval specified by **Compare point 1** and **Compare point 2**, then no change details are displayed.

6. Click **Compare**.
   The comparison view opens in a separate tab.

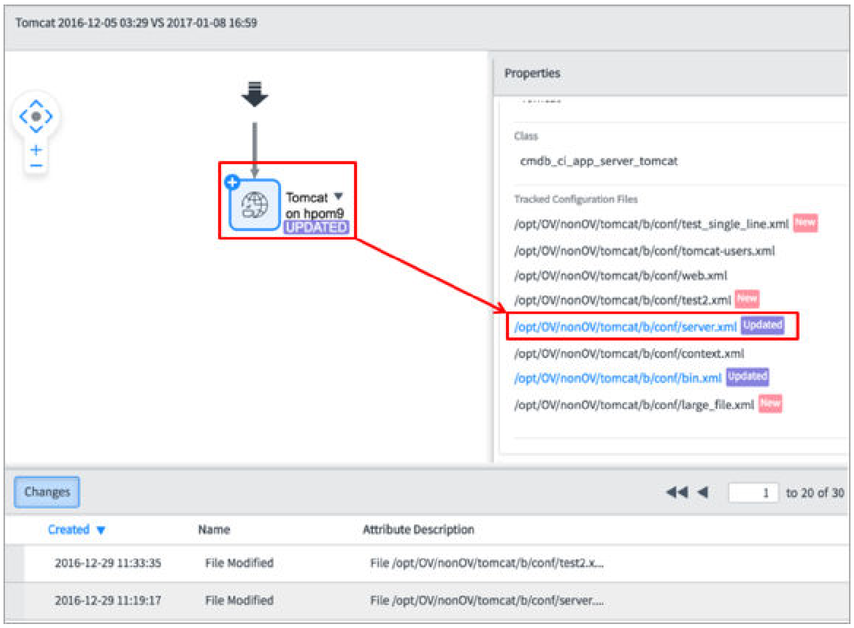
7. Select a marked CI to see the relevant change record on the **Changes** tab.



8. (Optional) If Service Mapping is deployed, you can compare two versions of a configuration file that appears on the map as Updated:

   a. Select the CI that is associated with the updated configuration file.

   b. In the **Properties** pane, click the link to the updated file.



The **Tracked Configuration Files Version Compare** tab opens showing two versions of the configuration file side by side.

c. Review actual changes.

   Highlight colors indicate the type of change:

   - Purple — Updated line
   - Pink — New line
   - Gray — Deleted line

d. Navigate between the changes using the Next difference icon and the Previous difference icon .

e. Close the **Tracked Configuration Files Version Compare** tab when finished.

9. Close the comparison view when finished.

Related tasks

- View the change history of application services in classic Service Mapping

Related topics

- Compare versions of CI configuration files

Use application services APIs

Application services provide APIs that let you perform operations such as creating and updating an service instance, populating it with CIs from the CMDB, and retrieving details from an existing service instance.

Role required: app_service_admin

A service instance is a set of interconnected applications and hosts that are configured to offer a service to the organization. Service instances can be internal, like an organization email system or customer-facing, like an organization website.

Create an application service

Using the createOrUpdateService - POST REST API to create an service instance suits your organization if the ServiceNow CMDB already contains the CIs making up the service. Typically, it is the case when you have manually added CIs directly into the CMDB, or used the Discovery application to discover CIs and store information about them in the CMDB. You can also use this API to create an service instance containing CIs discovered using non-ServiceNow applications.

By default, when an service instance is created, all CI connections are of the Depends on::Used by relationship type. You can modify this default type by changing the value of the sa.it_service.manual_ci_rel_type property.

Before creating an application service, ensure that:

- The CMDB contains all the CIs comprising the application service.
- You have the sys_id of each CI comprised in the application service you want to create.

- You understand the hierarchy that the CIs form.

The Mapped Application Service [cmdb_ci_service_discovered] table contains every service instance including services you create using APIs.

You can also manually create an service instance using the user interface as described in Create an application service.

Retrieve content from an application service

Use the getContent - GET REST API to retrieve a list of CIs and the relationships between them, for an application service that was created manually.

Additional APIs

The following JavaScript APIs are also available:

- addCI(): Add a CI to a manually created an service instance.

  For restrictions on the CIs being added and other details about adding a CI to an service instance, see Manually add CIs to an application service.

- addManualConnection(): Add a manually created connection to an application service.

- migrateManualToApplicationService(): Convert a manual service to an application service.

- populateApplicationService(): Populate an application service with CIs and relationships from the designated entry point.

- removeCI(): Remove a manually created CI from an application service.

- removeManualConnection(): Remove a manually created connection and the connected CI from an application service.

Related topics

- Application Service API

Components installed with application services

Several types of components are installed with activation of the Application Service [com.snc.cmdb.it_service] plugin, including tables, user roles, and scheduled jobs.

Note: The Application Files table lists the components that are installed with this application. For instructions on how to access this table, see Find components installed with an application.

Roles installed

Role title [name]	Description	Contains roles
[app_service_user]	Views maps for operational service instances and retrieves service content using the getContent - GET REST API. The itil role that serves as the basic helpdesk technician role contains the app_service_user role.	None
[app_service_admin]	Creates and modifies service instances, creates service	itil

Role title [name]	Description	Contains roles
	groups, views, and edits service instance maps.	

Tables installed

Table	Description
BaseLines [sa_baselines]	Storing points in the time defined as baselines for application services.
Business Service User preferences [sa_business_service_user_prefs]	User preferences associated with a specific application service.
Menu Action [sa_context_menu]	Data on configurable menu options for CIs in the application service map.
Hash [sa_hash]	Internal table which contains counters and hashes on various types of updates related to application services.
Entry Point [sa_m2m_service_entry_point]	Maps entry points to application services.
Discovered Service Notification [sa_notification]	Internal table which contains data on notifications between different parts of the software. Mostly used after activating Service Mapping.
Service Group Members	Maps service groups to application service members.

Table	Description
[sa_service_group_member]	
Business Service Group Responsibilities [sa_service_group_responsibilities]	Data on users having access to application service groups.
Checkpoint Attribute Description [checkpoint_attribute_description]	Links between history timeline changes and service model internal entities (checkpoints). Used in lists of history of changes in application service maps.
Service Instance [cmdb_ci_service_auto]	Services that can be monitored by the system, which in the base system, includes only application services. If Service Mapping is activated, there can also be records for dynamic CI groups. If Event Management is activated, there can be records for alert groups.
Mapped Application Service [cmdb_ci_service_discovered]	Application service CIs. For each application service, there is a container CI record that models the application service.
Bulk Convert Services [cmdb_convert_bulk_services]	All bulk conversions of business services to application services (current and past), along with the conversion progress which is refreshed every 10 seconds.
Application Services Action Results [csdm_dashboard_action_report_result]	Results for the 'Application Services Missing Data' report in the Application Services dashboard.

<table>
<tr><th>Table</th><th>Description</th></tr>
<tr><td>Application Services Types Results

[csdm_dashboard_type_report_result]</td><td>Results for the 'Application Services by Type' report in the Application Services dashboard.</td></tr>
<tr><td>Application Services Dashboard Results

[csdm_dashboard_reports_result]</td><td>Results for the '<Application Servers | Databases | Hardware Servers> Not in an Application Service' reports in the Application Services dashboard.</td></tr>
<tr><td>Manual CI Inclusions / Exclusions [svc_manual_ci_exclusions_inclusions.list]</td><td>Contains CI classes included or excluded from application services during population of manual or dynamic application services. Service population happens when
Manually adding CIs to an application service
Converting a business service to an application service
Creating or updating an application service using APIs
Manually updating an application service with changes from the CMDB
Note: The Manual CI Exclusions/Inclusions [svc_manual_ci_exclusions_inclusions] table doesn't include CIs added using traversal rules.

In the base system, the following CI classes are excluded:
NAT [cmdb_ci_translation_rule]</td></tr>
</table>

Table	Description
	• Endpoint [cmdb_ci_endpoint] • Qualifier [cmdb_ci_qualifier] • Application cluster [cmdb_ci_application_cluster] CIs of any CI class that is not configured for exclusion in this table can be added to application services. **Note:** In the CMDB hierarchy, a class derives the included/excluded setting from its closest ancestor. If a class has its own explicit setting, it overrides the derived setting. This table supports the functionality that was earlier supported using the following deprecated property: sa.mapping.user.manual.citype.blacklist.

Properties installed

To access application services properties, navigate to **All > Configuration > Application Services > Properties**. The role required for modifying property values, is app_service_admin.

If Service Mapping is deployed, see Properties installed with Service Mapping for additional application service properties.

Note: To open the System Properties [sys_properties] table, enter `sys_properties.list` in the navigation filter.

Property	Usage
The sys_id of the default relation type to be added between source and target when adding CI manually to application service sa.it_service.manual_ci_rel_type	• Type: string • Default value: 5599a965c0a8010e00da3b58b113d70e (Depends on::Used by) • Learn more: Manually add CIs to an application service
Coefficient of aggregation interval. 0 value means no aggregation is performed on history timeline. The purpose of this property is to decrease number of changes in history timeline by increasing the interval allowed between changes sa.history.aggr_interval_coef	• Type: integer • Default value: 1
Sync Service Mapping operations with Service Modeling sa.service_modeling.use	• Type: true | false • Default value: true
Enable limitation of application service maps drawing by number of nodes and edges. sa.map.LIMIT_MAX_GRAPH_SIZE	Limit the number of nodes and edges on application service maps. • Type: true | false • Default value: true

Property	Usage
	Setting this property to false may reduce performance in maps of large services.
Maximal number of displayable nodes on application service map. Maps with larger values will not be displayed. sa.map.MAX_NODES_FOR_LAYOUT	The max number of nodes displayed on an application service map. If the number of nodes exceeds the specified number, the map does not appear and an error message appears. • Type: integer • Default value: 5000
Global flag to allow or disable spanning tree view for maps. true (default) - allows but not forces spanning tree view on maps. sa.map.ALLOW_SPANNING_TREE_VIEW	Enable spanning tree view for application service maps. • Type: true | false • Default value: true
Maximal number of displayable edges on application service map before spanning tree view applied. sa.map.MAX_EDGES_FOR_FULL_LAYOUT	The max number of edges displayed on an application service map, before applying spanning tree view. • Type: integer • Default value: 1000
Maximal number of displayable edges on application service map.	Max number of edges displayed on an application service map. If the number of edges exceeds the

Property	Usage
Maps with larger values will not be displayed. sa.map.MAX_EDGES_FOR_LAYOUT	specified number, the map does not appear and an error message appears. • Type: integer • Default value: 100000 Increasing the default value may reduce performance in maps for large services.
Maximal degree of node on application service map for large map mode. Maps with smaller degrees will be displayed in regular mode. Maps with larger degrees will apply more edges merging for more compact view. sa.map.LIMIT_GRAPH_DEGREE	• Type: integer • Default value: 1000 Increasing the default value may reduce performance in maps for large services.
Limit of amount of services that displayed on Services Tree on maps. Then this limit reached, Services Tree will be blocked. sa.service_tree.MAX_ITEMS_TO_DISPLAY	• Type: integer • Default value: 7000
Maximal amount of connection properties to be shown at once when connection line selected on service map. If selected line contains more connections than defined here, then properties	• Type: integer • Default value: 50

Property	Usage
panel will have notification about cut-off connections. sa.map.max_connections_in_properties_panel	
Enable grouping of CIs on map. sa.map.enable_auto_grouping	• Type: true | false • Default value: true
Minimal number of CIs on a map to apply CI grouping. Relevant only if CI grouping is enabled on the map. The following CIs are not counted: discovered clusters, internal CIs inside inclusion boxes, entry points, error nodes, host CIs or CIs that are not hosted on other CIs. sa.map.min_nodes_for_auto_grouping	• Type: integer • Default value: 10

Render full labels on CIs on map. Applicable to all CI labels (CI name, host name, cluster label, etc.) Enabling this will disable labels truncation, and labels will most probably overlap with other map elements. Not applicable to network/storage path maps.
 	
sncCommonMap.RENDER_FULL_LABELS
	The default value of disabled, means none.
	• Type: choice list
	• Default value: Disabled
	• Other possible values:
	  ◦ Exported PDF only: pdf
	  ◦ Map and PDF views: all

Property	Usage

Maximal width of CI node labels in pixels. Relevant for any kind of labels (CI name, host name, cluster label etc.) This size also modifies horizontal space between CI elements. Applied to map view and exported PDF view. Not applicable to network/storage path maps.
 	
sncCommonMap.NODE_LABEL_WIDTH
	• Type: integer
	• Default value: 95
	• **Other possible values**:
	  • Min value: 20
	  • Max value: 1000

glide.cmdb.csdm.app_service.max_results	Max number of items that are calculated in the '<Application Servers | Databases | Hardware Servers> Not in an Application Service' report in the Application Services dashboard. • Type: integer • Default value: 100000 • Location: Add to System Properties [sys_properties] table.
sa.service.max_ci_service_population	The maximum number of CI connections added to application services during the following operations: Converting manual services created in Event Management into application services and updating application services with changes from the CMDB. • Type: integer

Property	Usage
	• Default value: 1,000 • Location: Add to System Properties [sys_properties] table. Increasing the default value may cause performance issues.
sa.service.population.stop_expansion_under_ci_classes	List of application service CI classes. If an application service belongs to a CI class that extends one of the CI classes in the list, the system does not insert CIs under this application service CI during Manually updating an application service with changes from the CMDB. • Type: string • Default value: cmdb_ci_service_discovered • Location: System Property [sys_properties] table.